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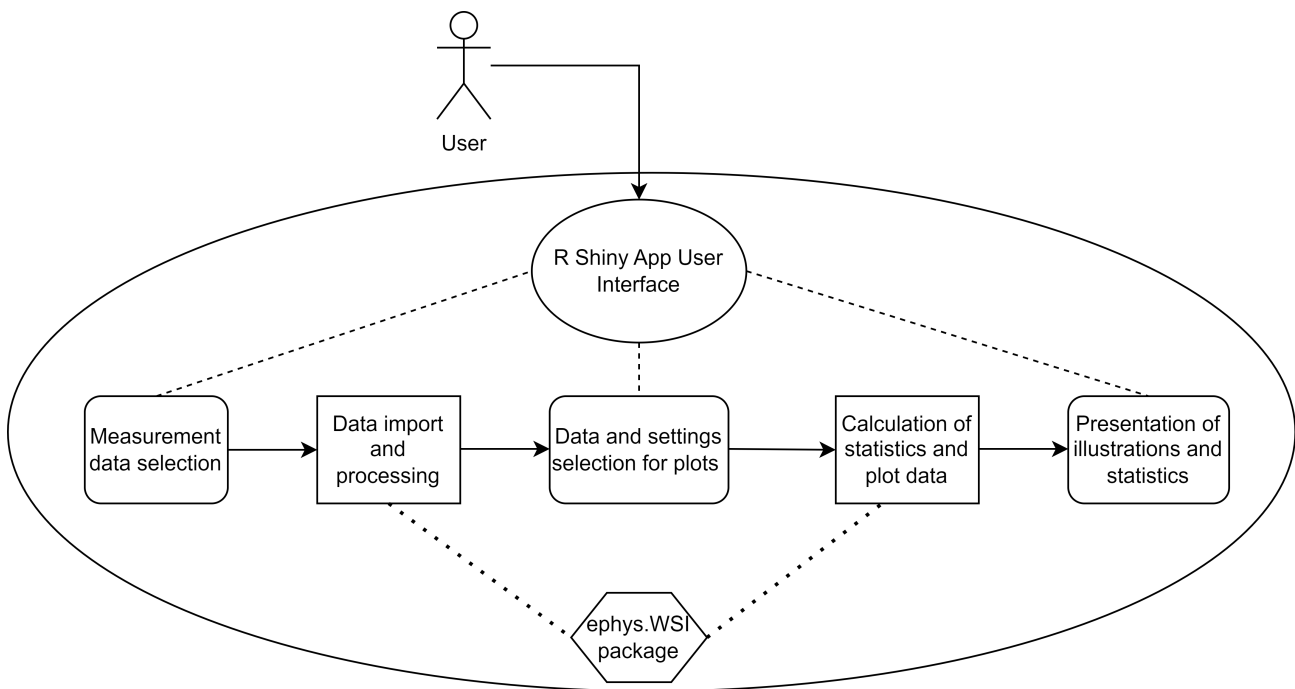
1_Home.md

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Data-Analysis.md

Overview



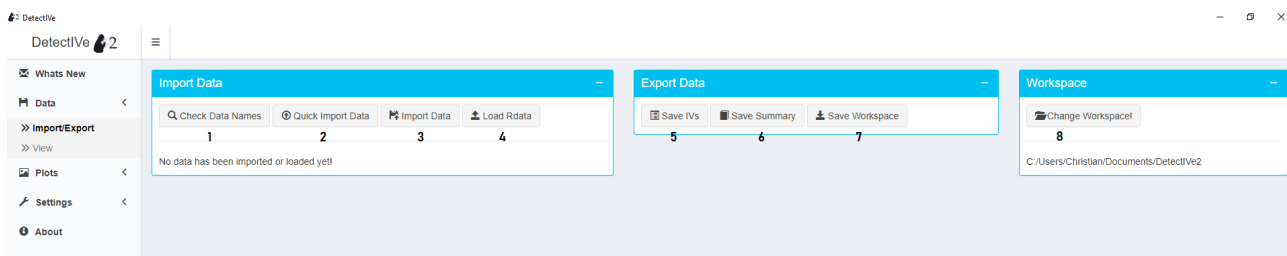
Preparation

HEKA EPC10 Patchmaster voltage ramp sweeps need to be exported as `.asc` files (IV). Each file needs a corresponding entry in a summary `.xlsx` file, which contains the file name, the cell capacity and the number of the exported sweep. Different data sets can be combined in the summary `.xlsx` files sheets.

Internal Workflow

1. The median values of the currents at the [holding potential](#) are calculated.
2. Any unnecessary columns are removed from the `.asc` files and the data is cleaned up.
3. The `ASC` files and summary are linked manually.
4. The holding potential median values are written to the summary.
5. The offset is determined by subtracting the value at the reverse potential in the IVs from 0. This value is then written in the summary.
6. The current densities are calculated and written in the IVs and summary.
7. The current densities are offset-corrected and written in IVs and the summary.
8. The current densities inward/outward ratios are calculated and written in the summary.
9. The [normalized slope conductance](#) (00226-0) is calculated and written for IVs.
10. Outliers are identified using the iterative Grubbs method and are included in the summary.

Data-Import-and-Export.md



Data Import

1. Use this function to make sure your data is free of easy to detect errors such as typing errors or missing .asc files. Mimics the Quick Import Data(2) function.
2. This is the fastest and most convenient way to import data. After selecting a summary .xlsx file the import process starts automatically.
Important: The .asc files **must** be stored in a folder with the same names as the sheet of .xlsx file they are referenced in, in a folder called *ASC files* in the selected workspace.
3. Another way to import data. After selecting a summary .xlsx the user needs to select manually all .asc files that are mentioned in the summary .xlsx.
4. If previously imported data has been saved as a .rdata file it can be easily reloaded.

Throughout the import process the different imported IVs need to be characterized and linked to the summaries. This is done through the peak setup and the column setup.

Data Export

5. Saves the adjusted and extended (current densities, NSC etc.) .asc files.
6. Saves the adjusted and extended (current densities, ratio, outlier etc.) .xlsx file.
7. Saves the whole imported data as .rdata file for easy reusing.

Workspace

8. Here the workspace can be changed. The workspace determines the directory where plots will be saved and also where import and export directory selection windows start.

View

The view tab can be used to get a quick or longer look at the imported data.

Detective

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Whats New

Data

Import/Export

View

Plots

Settings

About

Selection

Choose Series

mTRPC5

Choose IV

mTRPC5_IRES2_EGFP_BTDAzo(10uM)_S1_1(3)_BTDAzo

Summary

Show 10 entries

Search:

	Name	Cm	SweepBef	Iminbef	Imaxbef	SweepBTDAzo	IminBTDAzo	ImaxBTDAzo	ID	Offsetbef	OffsetBTDAzo	CD_Im
1	mTRPC5_IRES2_EGFP_BTDAzo(10uM)_S1_1(3)	9.48	54	-127.612834291	431.7028531885	84	-2106.8848288	3499.234140195	1(3)	16.89076136	134.500508	-13.46126
2	mTRPC5_IRES2_EGFP_BTDAzo(10uM)_S2_2(3)	10.5	33	-131.141126779	1370.86988701	70	-3296.460547865	5202.026249770001	2(3)	11.26050815	190.177471	-12.48963
3	mTRPC5_IRES2_EGFP_BTDAzo(10uM)_S3_3(3)	7.18	15	-31.05085089255	543.8825373435	40	-2366.29252634	5383.76773382	3(3)	2.502335145	115.1074167	-4.3246310
4	mTRPC5_IRES2_EGFP_BTDAzo(10uM)_S4_4(3)	9.43	8	-173.552579317	1546.840346345	53	-3346.27889509	7085.061579655	4(3)	10.00934058	205.1914749	-18.404300
5	mTRPC5_IRES2_EGFP_BTDAzo(10uM)_S5_5(3)	7.54	34	-74.3819108912	699.521522652	61	-3549.64992956	6197.498978405	5(3)	5.630254075	272.1289305	-9.8649748
6	mTRPC5_IRES2_EGFP_BTDAzo(10uM)_S6_6(3)	3.25	70	-67.30343067615	435.112284121	96	-1691.372091785	6292.243621675	6(3)	6.881421648	75.69563726	-20.708742
7	mTRPC5_pIRES2_eGFP_BTDAzo(10uM)_S1_1(2)	8.2	46	-207.7111209455	745.749070551	74	-3668.87613624	3909.016035315	1(2)	4.378462436	40.65714779	-25.330624
8	mTRPC5_pIRES2_eGFP_BTDAzo(10uM)_S2_2(2)	11.05	14	-103.4974646571	413.764681126	35	-2306.69275013	4399.866659895	2(2)	10.63340822	-43.15912905	-9.366286

Showing 1 to 10 of 11 entries

Previous12Next

IV

Show 10 entries

Search:

	Index	Time[s]	CurrentIn[A]	Potential[mV]	Offset	CurrentIn[A]_OffsetCor	CurrentDensity[pA/pF]	CurrentDensity[pA/pF]_Inward	CurrentDensity[pA/pF]_Outward	smoothed_CurrentDensity	snr
1	250	0.05	-2.061924187e-9	-100	1.34500508e-10	-2.196424695e-9	-231.6903686708861	-231.6903686708861		-228.8218721285923	
2	251	0.0502	-2.055668302e-9	-99.9	1.34500508e-10	-2.19016881e-9	-231.0304651898734	-231.0304651898734		-228.5972290805244	
3	252	0.0504	-2.061298465e-9	-99.8	1.34500508e-10	-2.195798973e-9	-231.6243642405063	-231.6243642405063		-228.3725840346645	

Installation.md

Requirements

Hardware

About 1GB of free space.

Software

R 4.2.3 must be installed. [Download from CRAN](#)

Java 64 bit version must be installed. [Download from Java.com](#)

Installer

Use [Inno Setup](#) to create an executable installer from the R project using the *rlInno.R* script followed by the *DetectIve2.iss*.
[Node.js](#) might be needed to succesfully create the installer.

Other

It is recommended to use a local harddrive.

The first startup might take some minutes depending on the used computer

Main-Menu.md



The main menu can be accessed through the sidebar and be used to select the wanted functionalities.

- Data: Import/Export and view the data
- Plots: Create different plots
- Settings: Change the settings for data import and plot creation

Plots.md

General Settings

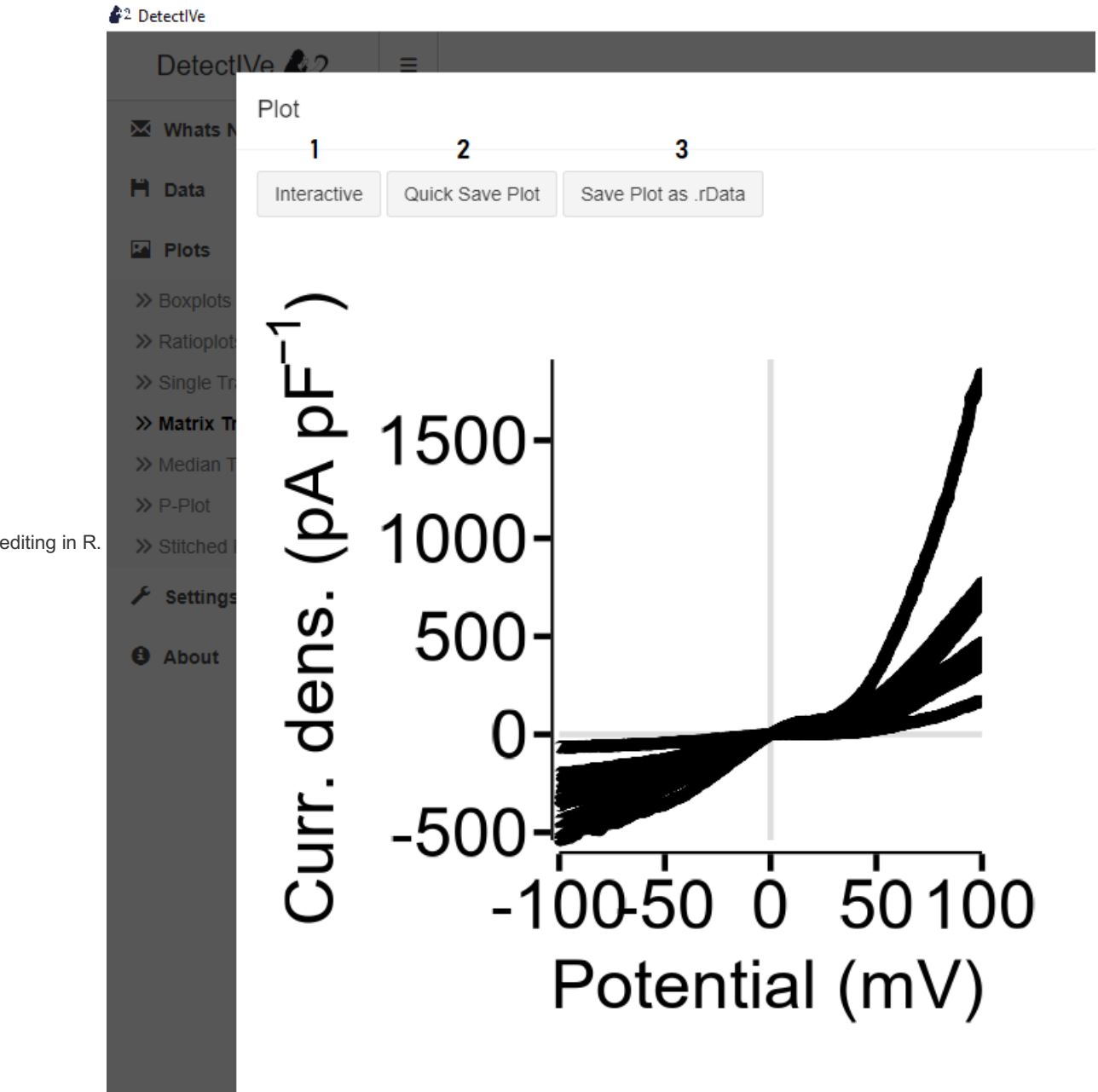
The settings for saving a plot, changing the colors, and displaying a plot are the same for all different kinds of plots.



The Save Plot box (1) is used to save the plot with a name and a file type of your choosing. The width and height of the plot are given in millimeters. Once the Save Plot button is pressed, the plot will be saved in the current workspace.

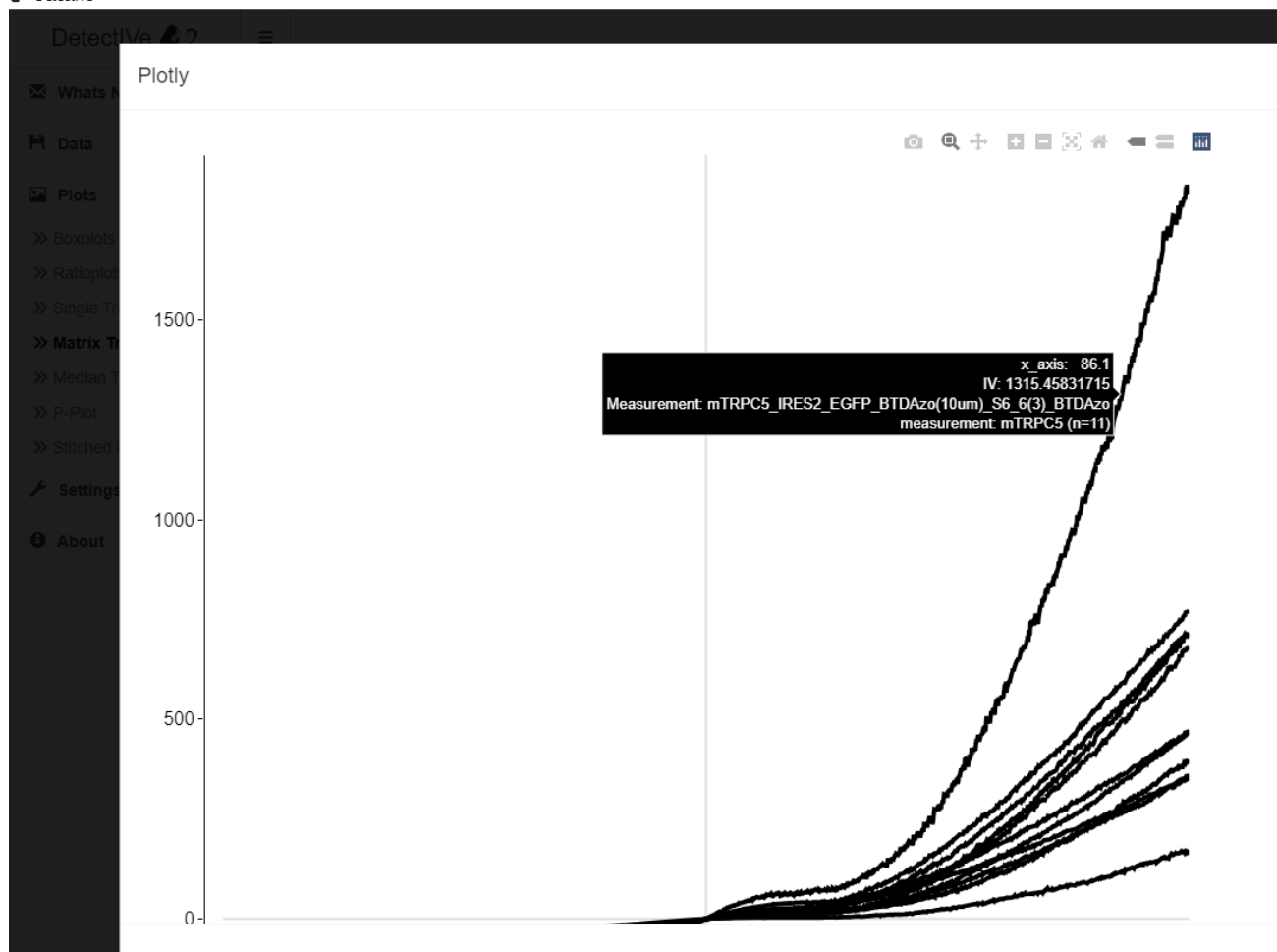
The Colors box sets up the colors for the different plot elements. Colors are given as HTML color codes or hexadecimal codes. Either individually (a) or as text separated by semicolons (b). Different color inputs can be viewed and chosen through the color list (c). The switch (_d_) is used to change between the different color inputs (_a_ & _b_). Hexcodes for different colors can be found on websites such as [htmlcolorcodes](http://htmlcolorcodes.com).

Through the Plot Updates (_3_) box the active plot can be either viewed as size reduced preview plot (_e_) in the preview box or in fullsize in a modal window (_f_). The modal plot window includes the created plot and 3 buttons to make the plot an interactive html plot (_1_), to quicksave the plot in the current workspace (_2_) and to save the plot as .rData file (_3_) for later

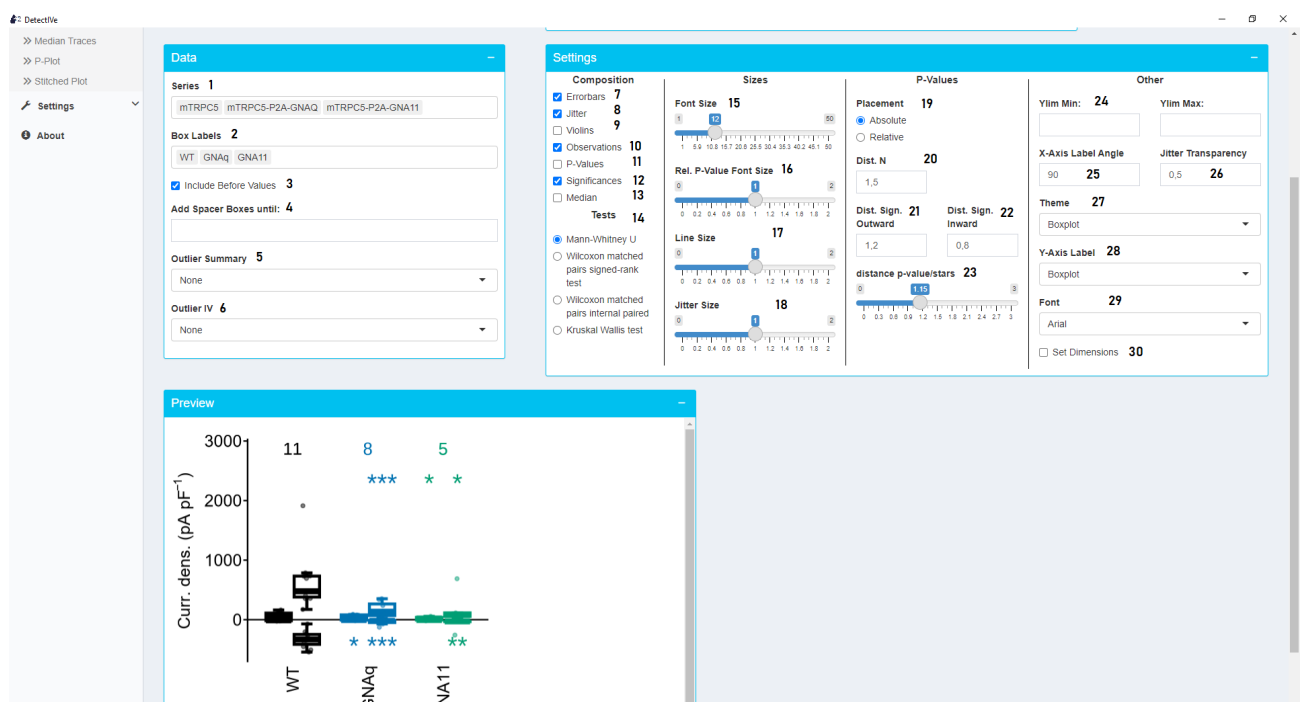


The interactive plot can be used to analyze plots in more detail, by allowing the user to zoom in order to get extra infos when hovering over specific parts of the plot.

2 DetectIVe



Boxplots

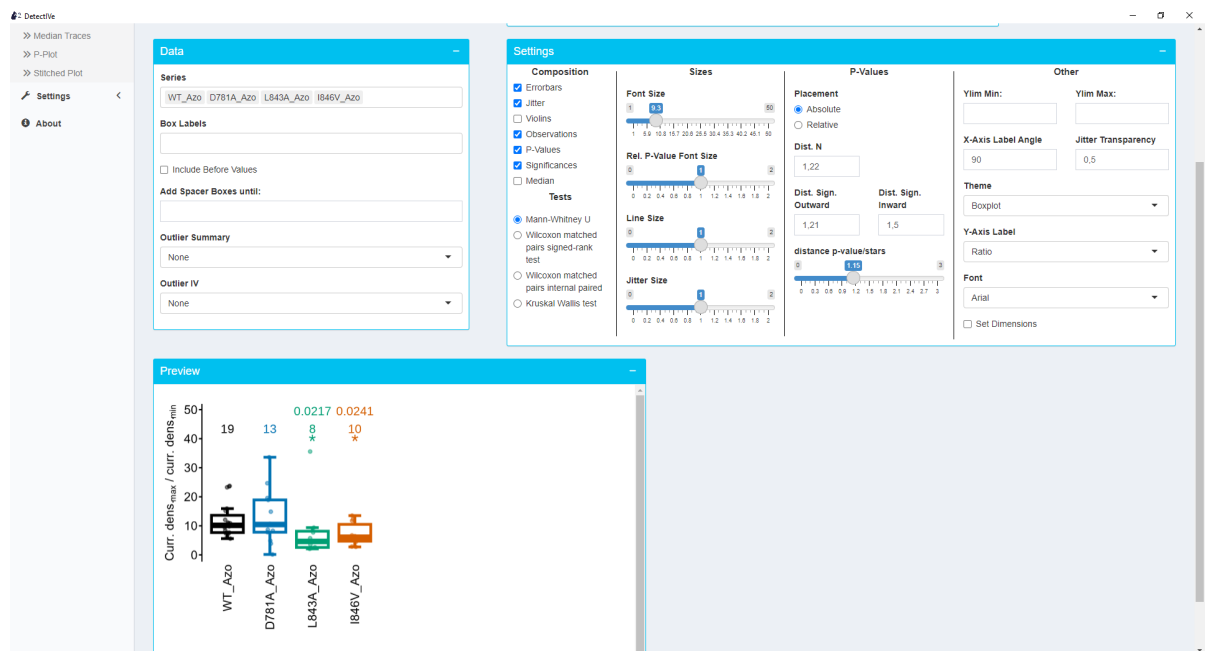


- Data
 - Select which series to display as box plots (_1_) and how to label them under the X-axis (_2_).
 - Choose whether to display the "Before/Bef" boxes (_3_).
 - Adding spacer boxes (_4_), which can be removed later by editing the graph, can be useful to obtain boxes of the

same size when creating several plots with different number of series.

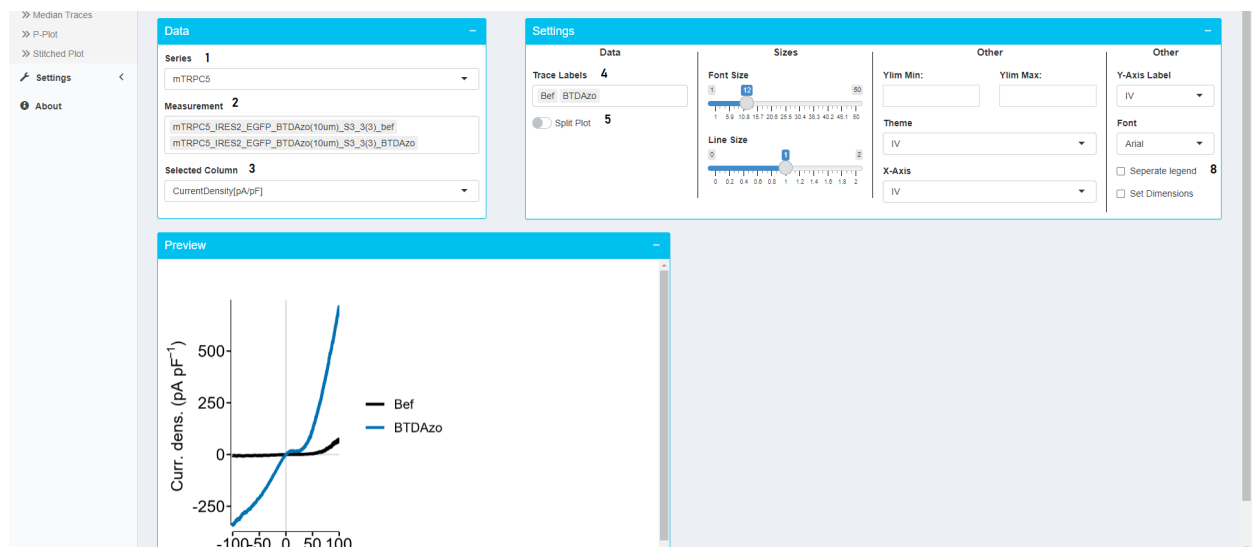
- Select whether outliers should be part of the plot or not (_5_&_6_).
- Composition
 - Add or remove the error bars from the boxes (_7_).
 - Add or remove the jitter/dots from the boxes (_8_).
 - Add or remove violin plots (_9_).
 - Add or remove the number of observations (_10_).
 - Add or remove the p-values of the selected significance test (_11_).
 - Add or remove the stars corresponding to the p-values of the significance test (_12_).
 - Add or remove the display of the median +/- standard deviation (_13_).
- Tests
 - Change the statistic test which will be used to test data (_14_).
- Sizes
 - Change the font size of all text elements in the plot (_15_).
 - Change the size of the displayed P-Values relative to the other text elements (_16_).
 - Change the line size (_17_).
 - Change the jitter point size (_18_).
- P-Values
 - Change the placement type (_19_) either to *Absolute* (All extra info elements will be at the same height) or to *Relative* (All extra info elements will be placed above the individual boxes).
 - Change the distance between the top of the boxes and the observation count (_20_).
 - Change the distance between the top/bottom of the boxes and the significance stars (_21_&_22_).
 - Change the distance between the significance stars and the p-values (_23_).
- Other
 - Set the limits of the y-axis (_24_).
 - Set the angle of the x-axis labels (_25_).
 - Set the transparency of the jitter points (_26_).
 - Set the design for the plot. It is recommended to use the theme with the same name as the plot for maximum compability. The analysis themes are giving extra tools such as gridlines (_27_).
 - Set the label of the Y-Axis. Use one of the prewritten templates or use a custom one (_28_).
 - Change the used font (_29_).
 - Use fixed plot dimension (_30_).

Ratioplots

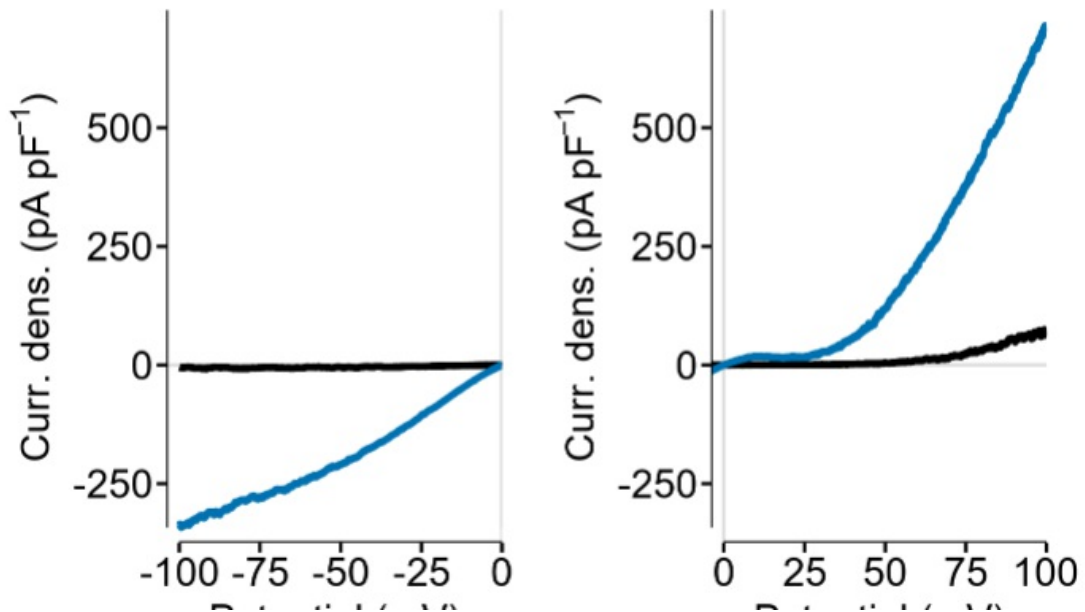


- Settings for ratioplots are identical to boxplots.

Single Traces

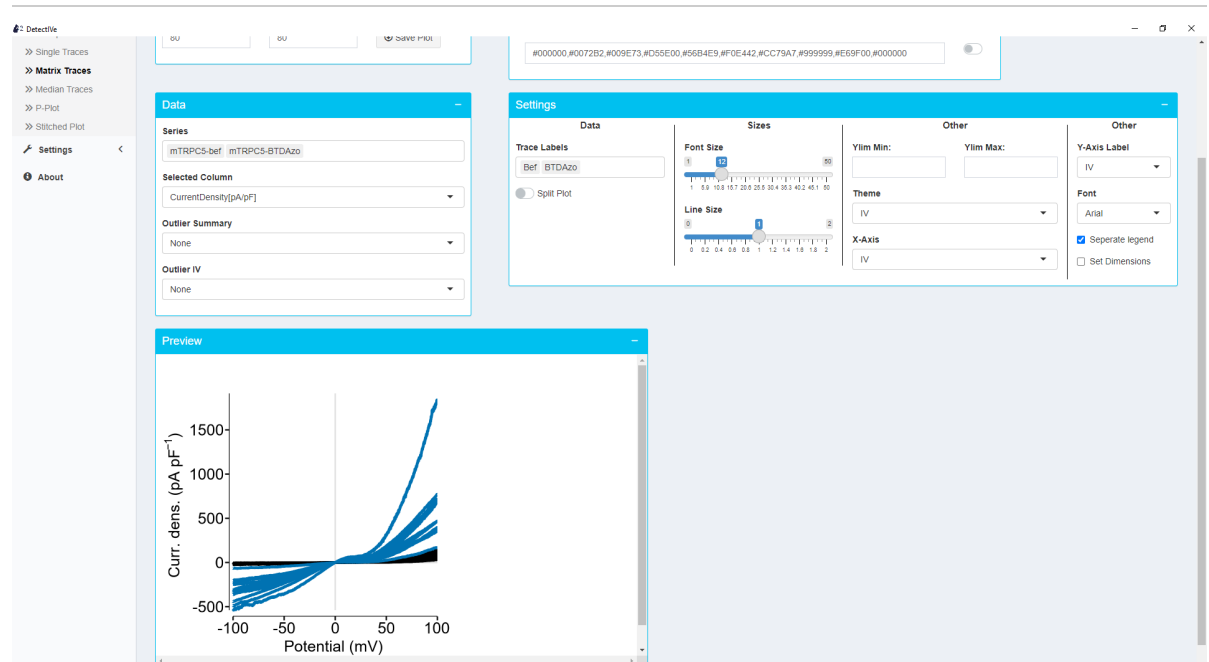


- Data
 - Select the series (summary sheet) which contains the data to be displayed (_1_).
 - Select the measurements to be displayed (_2_).
 - Select the measurement values to be displayed (_3_).
- Settings
 - Change the legend labels of the selected measurements (_4_).
 - Display the plot as whole or splitted at the reverse potential (_5_).



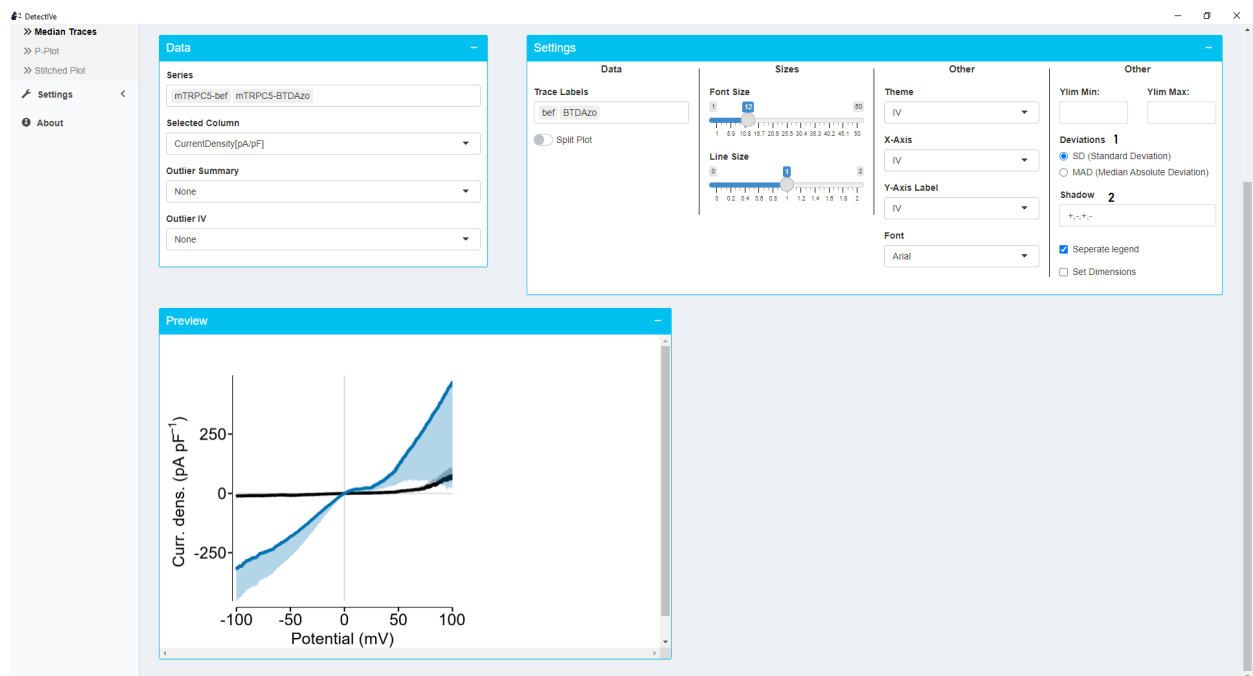
- Separate the legend from the plot and save it as individual file (_8_).

Matrix Traces



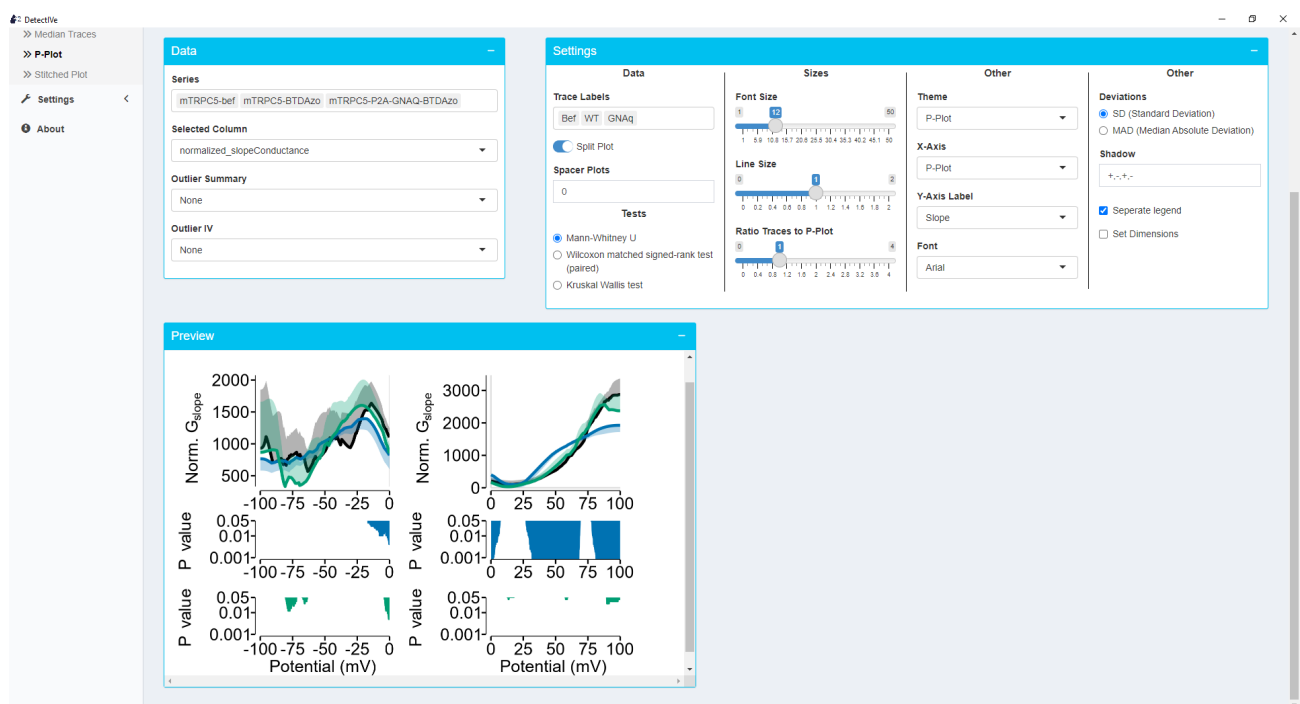
- Settings are identical to previous plots

Median Traces



- Settings
 - Select the way the displayed deviations are calculated (_1_).
 - Select the direction the deviation shadow will be displayed (+ for on top of the line; - for underneath the line) (_2_).
- Other options are identical to previous plots.

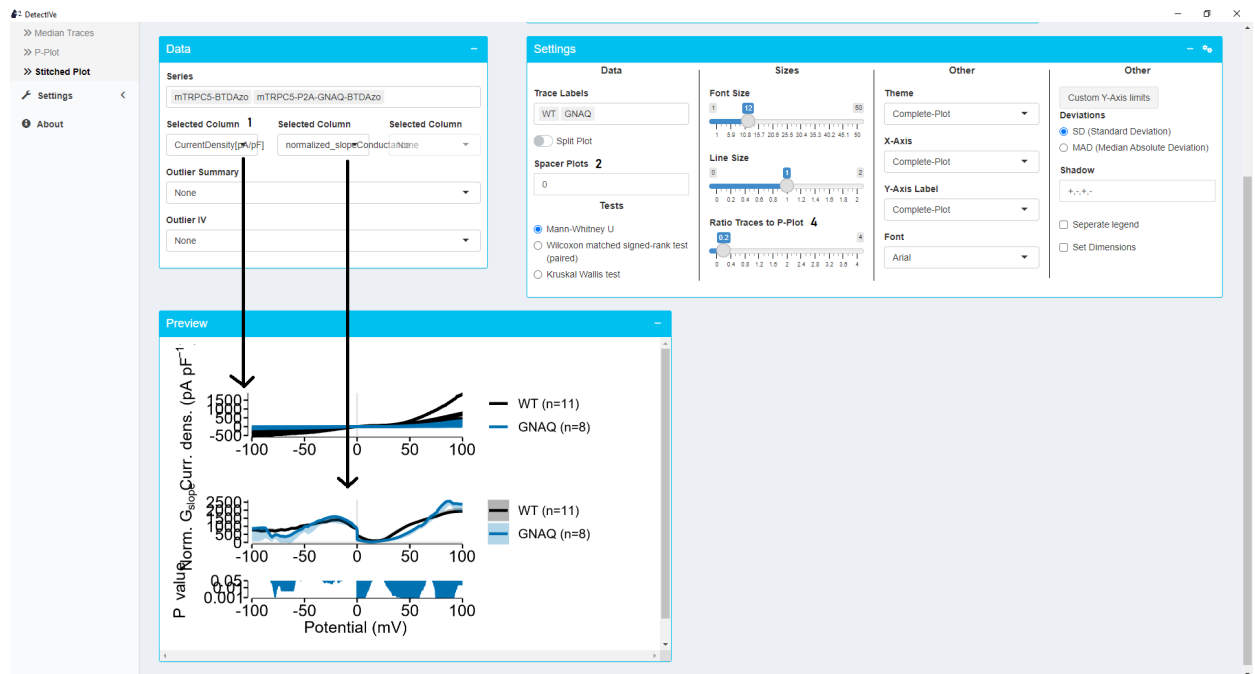
P-Plot



- Settings are identical to previous plots.

Stitched Plots

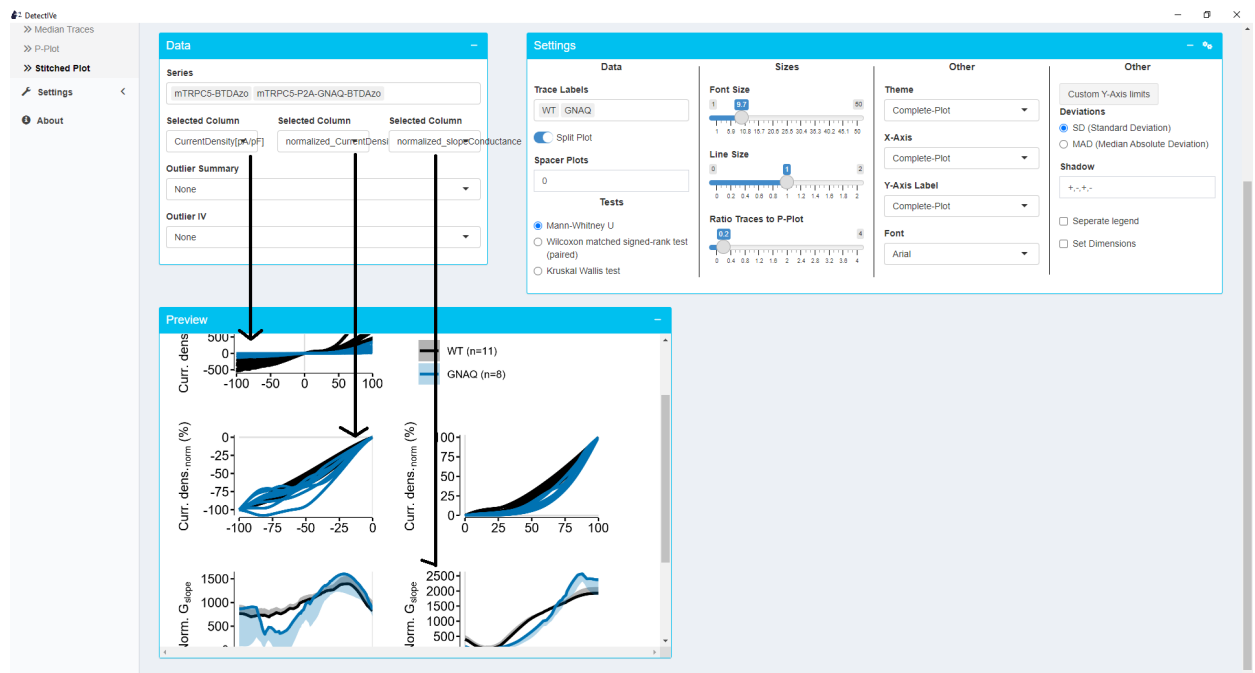
Stitched Plots complete



- Data
 - Select the columns which will be displayed (_1_). The arrows indicate the position in the final plot.
- Settings
 - Adding spacer plots(_2_), which can be removed later by editing the graph, can be useful to obtain plots of the same size when creating several plots with different number of series.

- All remaining options are identical to previous plots.

Stitched Plots splitted



Settings.md

Import

It is possible to change various settings in order to customize data import processes.

1. Current Export:

The index markers (`_b_` & `_c_`) are used to extract the median values of the holding potentials. The red lines in `_d_` indicate the sections that have been used. There are preset configurations for the most commonly used protocols in our laboratory (`_a_`).

2. Split at reverse potential:

If the option is selected, the IVs and all parameters dependent on the IV, such as the NSC, will be divided at the specified reverse potential. For instance, all currents for a potential range from -100 V to 0 V and all currents for a potential range from 0 V to +100 V will be divided.

3. Fit through reverse potential:

Should this option be selected, the smoothing function used for NSC generation will be forced to go through the reverse potential. This will result in more visually appealing plots, but may affect the accuracy and truth of the data.

4. Fit parameters:

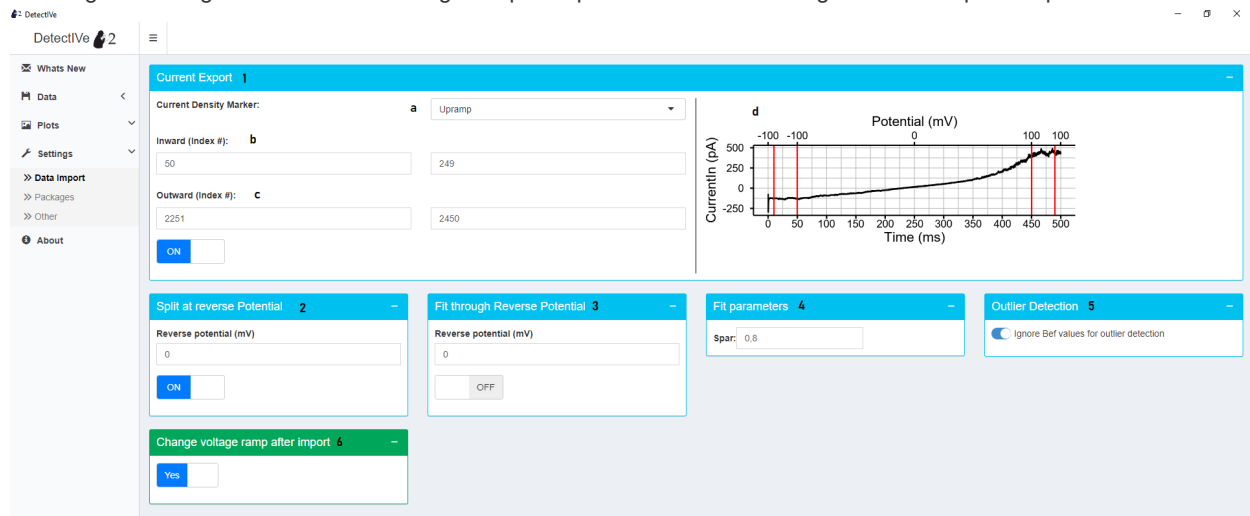
This option can be used to change the fit parameter `spar` of the smoothing function.

5. Outlier detection:

If set to on, the before (`bef`) values will not be used for outlier calculation.

6. Change voltage ramp after import:

Enabling this setting ensures that the voltage ramp of imported data will be changed to the ramp date specified in [Other](#)



Packages

Please note that this page is not required in the installed version.

This page displays the installed versions of the *ephys.WSI* package, as well as the version of all necessary support packages to run without errors. In the event that packages are missing, they can be installed via the 'Install' button at the bottom.

Other

These settings handle mainly the voltage ramps.

1. The voltage ramp characteristics are entered here. Changing the [current export configuration](#) will also change the preset data. If a different ramp is to be used, it is necessary to manually update the ramp data and confirm it using the change button before importing the data.
2. If [Change voltage ramp after import \(6\)](#) is set to on, the voltage ramp will be converted to the properties of this box through the data import process.

Detective 2

What's New

Data

Plots

Settings

» Data Import

» Packages

» Other

About

Original Voltage Ramp 1

Ramp Start: -0,1

Ramp End: 0,1

Ramp Length: 2001

Ramp Steps: 0,0001

Ramp Unit: V

Change

New Voltage Ramp 2

Ramp Start: -100

Ramp End: 100

Ramp Length: 2001

Ramp Steps: 0,1

Ramp Unit: mV

Manually Change Voltage Ramp 3

Ramp Start: -100

Ramp End: 100

Ramp Length: 2455

Ramp Steps: 0,1

Ramp Unit: mV

Switch Voltage Ramp

Plot Dimensions 4

Boxplot

Upload Dimensions

Save Dimensions

- This settings can be used to change the voltage ramp of already imported data manually.
- This option can be used to extract and upload the dimensions of a plot. Can be usefull to make plots with the same dimensions and margins. (Might not worked in the installed version)